

TIM 172 A, MOT-I, Lecture #13 (11/06/25)

Agenda:

1. Review L #12
2. New Product Conceptual Design Process
↓
3. HW #6 (for you to work on in the coming week)

1. Review of L # 12

(a) Project Planning for complex projects

- posted L # 12 notes on Canvas
- Read the chapter "Managing Projects" in the PD & D text by U & E.

(b) 3 important ideas for conceptual design

- function structure (FS)
- dissecting related products to generate most of the sub-functions for the FS for your new product.
- add new sub-functions to obtain the complete FS for your new product.

2. Conceptual Design of New Products

Statement of the problem(s):

1. Generate several alternative concepts to realize (make real) a high-level idea for a new product (that satisfies real customer needs, i.e. customer centric.)
2. Compare and assess these concepts relative to each other; and rank them.
3. Select one (or two) feasible concepts for further development

The process (or plan or approach) for new conceptual design:

The first three steps are done concurrently, or in parallel

Step 1 : Identify the primary function of the product (aka the product INTENT)

Example : Autonomous system to perform
(high-level) useful tasks indoors
(HW #6, Prob. 2)

Step 2 : Establish customer needs & technical metrics (or specifications) for the new product using the HOQ.

Step 3 : Identify one or more related products, & then reverse engineer (or dissect) these products using FAST.

Example : Dissect related products
(HW #6, Prob. 2) such as the ROOMBA,
⇒ identify important sub-functions of these products from the FAST diagrams.

(Note : In order to identify related products, it is useful [important] to create a functional map showing the evolution of products in the "space" in which the new product is embedded.)

Step 4 : Create an abstract functional representation of the new (intended) product, called the Function Structure (FS)

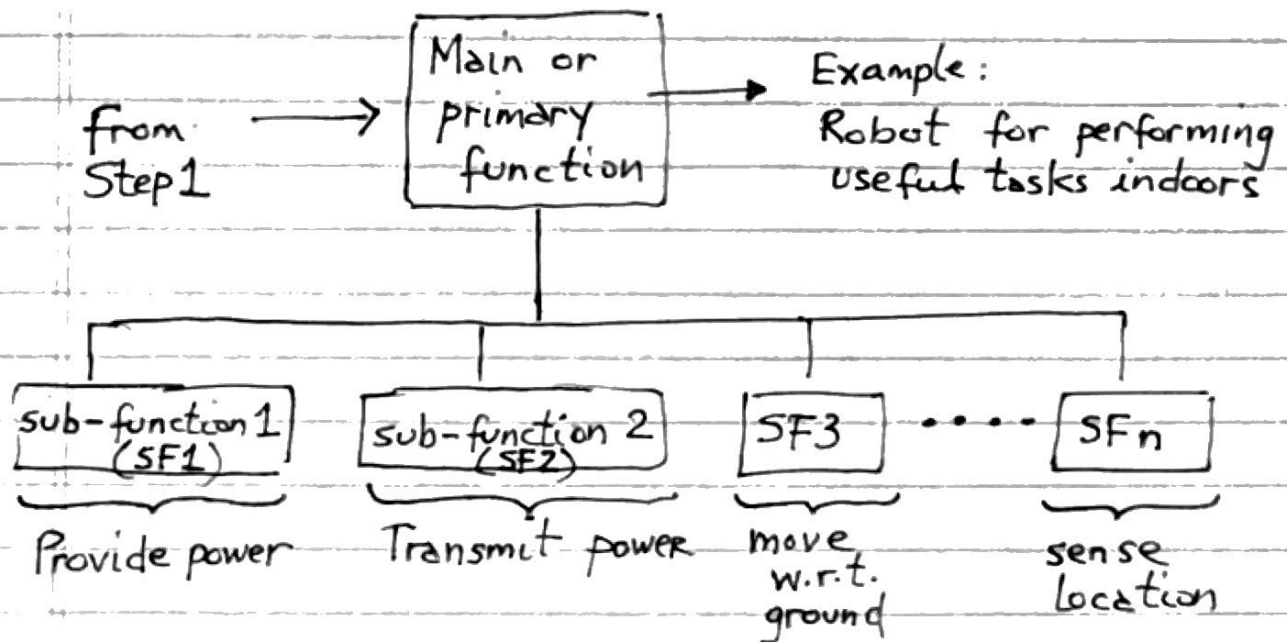
FS is essentially a FAST diagram without the realizations (or sub-systems or FORM or HOW?)

$FS \equiv$ "Whys" only

Example : Function Structure (FS) for the autonomous indoor system

Remember : we have dissected related products, from which we can extract relevant sub-functions : e.g., (in this case) Roomba,...

FS for autonomous indoor system :



(Note : sub-functions correspond to WHY?)

Step 5:

(i) for each sub-function, generate solution-principles (SPs) for realizing (i.e. making real) that sub-function.

(Note : SPs correspond to HOWs?)

(ii) Organize the sub-functions & solution principles (SPs) into a morphological matrix (MM)

[due to Zwicky]

Example : Morphological Matrix (MM) for the autonomous indoor system

Solution principles (SPs) → sub-functions (SFs) ↓	Solution principle 1 (SP1)	SP2	SP3	
SF1: (provide power)	Electricity	Gasoline	Solar	Wind
SF2: (transmit power)	Gearbox	Hydraulic	Pneumatic (air)	
SF3: (move w.r.t. ground)	wheels	legs	track (thread)	air cushion
SF4: (sense location)	Vision	Sonar	GPS	
...	①	②	③	

all other sub-functions (HW#6, Prob. 2)

Step 6 : Generate 6-10 alternative concepts by suitably combining the solution principles in the MM.
(see above)

Note : The first few columns of the MM (Nuance) should typically contain the "form" (SP) information for the dissected related products.

Example : column 1 of the MM would (in this case) contain all the form (SP) information for the ROOMBA

Very important : Each of the conceptual design alternatives must be sketched, explained in words, & presented as a FAST diagram

Step 7 : Identify an appropriate set of selection criteria to compare/assess the alternatives from Step 6; and use these criteria to create a utility function (UF)

This utility function will be used to compare, rank, and select from these alternative concepts (see L #14).

Step 8 : Use the UF to select 1-2 feasible concepts for further development.